

#1

The screenshot shows a database IDE with a dark theme. The top panel displays a SQL script for creating a procedure and calling it. The left sidebar shows a database explorer with a tree view of the database structure. The bottom panel shows a table view of the 'flights' table with three rows of data.

**SQL Script:**

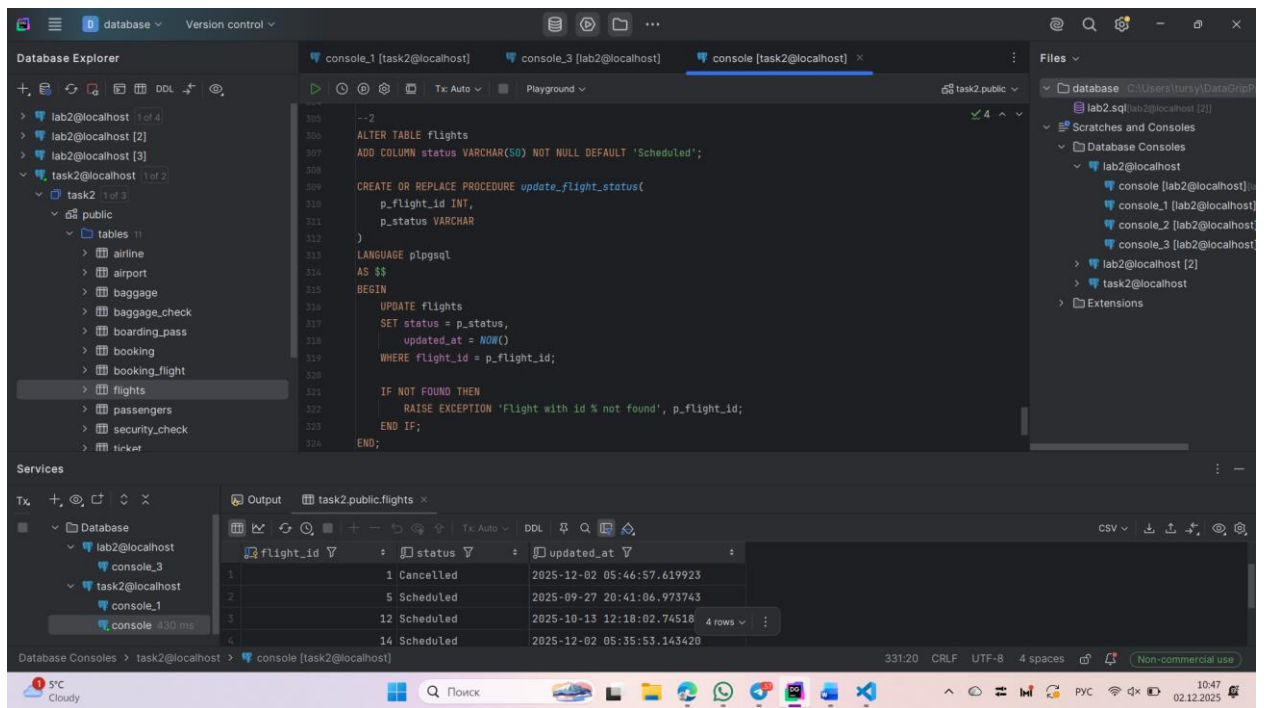
```
CREATE OR REPLACE PROCEDURE insert_flight(  
    p_sch_departure_time TIMESTAMP,  
    p_sch_arrival_time TIMESTAMP,  
    p_departing_airport_id INT,  
    p_arriving_airport_id INT,  
    p_departing_gate TEXT,  
    p_arriving_gate VARCHAR,  
    p_airline_id INT  
)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    INSERT INTO flights(  
        sch_departure_time,  
        sch_arrival_time,  
        departing_airport_id,  
        arriving_airport_id,  
        departing_gate,  
        arriving_gate,  
        airline_id,  
        act_departure_time,  
        act_arrival_time,  
        created_at,  
        updated_at  
    )  
    VALUES (  
        p_sch_departure_time,  
        p_sch_arrival_time,  
        p_departing_airport_id,  
        p_arriving_airport_id,  
        p_departing_gate,  
        p_arriving_gate,  
        p_airline_id,  
        p_sch_departure_time,  
        p_sch_arrival_time,  
        NOW(),  
        NOW()  
    );  
END;  
$$;  
  
CALL insert_flight(  
    '2024-02-15 10:00',  
    '2024-02-15 14:00',  
    1,  
    2,  
    'A5',  
    'B3',  
    1  
);
```

**Table View: task2.public.flights**

| flight_id | sch_departure_time         | sch_arrival_time           | departing_airport_id | arriving_airport_id | departing_gate |
|-----------|----------------------------|----------------------------|----------------------|---------------------|----------------|
| 14        | 2024-02-15 10:00:00.000000 | 2024-02-15 14:00:00.000000 | 1                    | 2                   | A5             |
| 12        | 2024-01-10 08:01:00.000000 | 2024-01-10 12:01:00.000000 | 1                    | 2                   | A1             |
| 5         | 2024-02-10 08:00:00.000000 | 2024-02-10 12:00:00.000000 | 1                    | 201                 | A1             |

```
CREATE OR REPLACE PROCEDURE insert_flight(  
    p_sch_departure_time TIMESTAMP,  
    p_sch_arrival_time TIMESTAMP,  
    p_departing_airport_id INT,  
    p_arriving_airport_id INT,  
    p_departing_gate TEXT,  
    p_arriving_gate VARCHAR,  
    p_airline_id INT  
)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    INSERT INTO flights(  
        sch_departure_time,  
        sch_arrival_time,  
        departing_airport_id,  
        arriving_airport_id,  
        departing_gate,  
        arriving_gate,  
        airline_id,  
        act_departure_time,  
        act_arrival_time,  
        created_at,  
        updated_at  
    )  
    VALUES (  
        p_sch_departure_time,  
        p_sch_arrival_time,  
        p_departing_airport_id,  
        p_arriving_airport_id,  
        p_departing_gate,  
        p_arriving_gate,  
        p_airline_id,  
        p_sch_departure_time,  
        p_sch_arrival_time,  
        NOW(),  
        NOW()  
    );  
END;  
$$;  
  
CALL insert_flight(  
    '2024-02-15 10:00',  
    '2024-02-15 14:00',  
    1,  
    2,  
    'A5',  
    'B3',  
    1  
);
```

#2



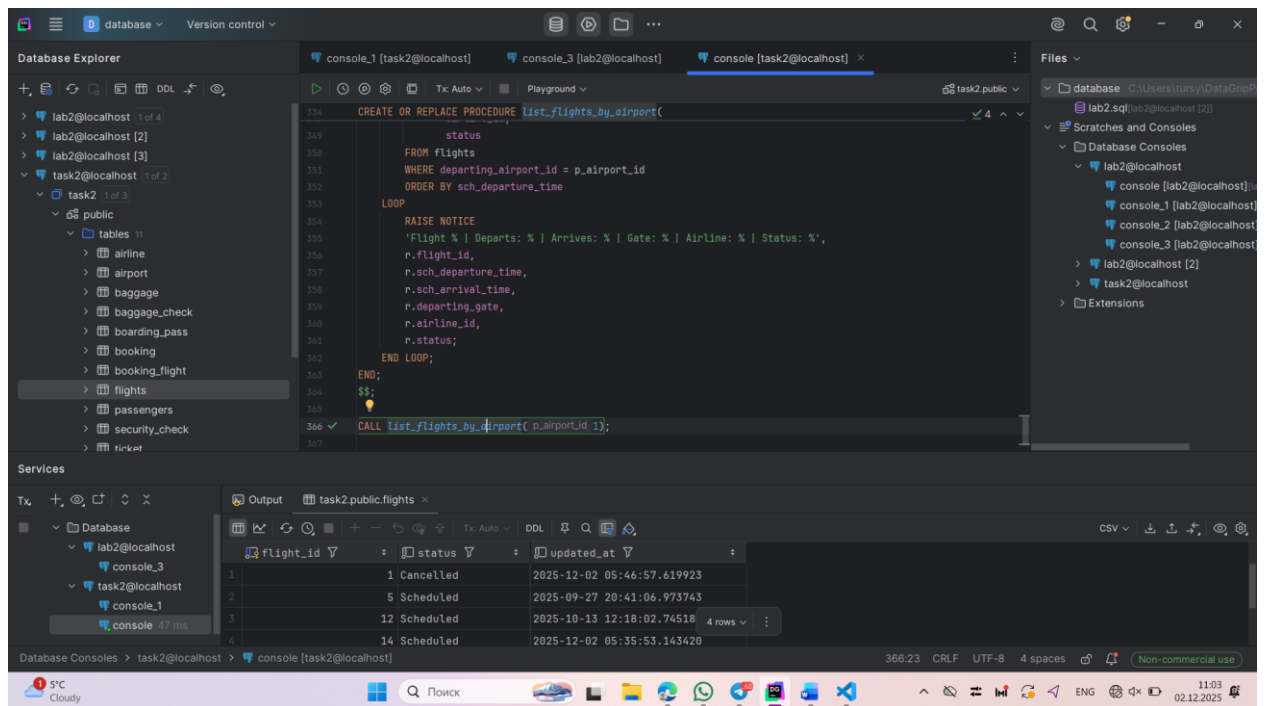
```
ALTER TABLE flights
ADD COLUMN status VARCHAR(50) NOT NULL DEFAULT 'Scheduled';
```

```
CREATE OR REPLACE PROCEDURE update_flight_status(
    p_flight_id INT,
    p_status VARCHAR
)
LANGUAGE plpgsql
AS $$
BEGIN
    UPDATE flights
    SET status = p_status,
        updated_at = NOW()
    WHERE flight_id = p_flight_id;

    IF NOT FOUND THEN
        RAISE EXCEPTION 'Flight with id % not found', p_flight_id;
    END IF;
END;
$$;
```

```
CALL update_flight_status(1, 'Delayed');
CALL update_flight_status(1, 'Cancelled');
SELECT flight_id, status, updated_at
FROM flights
ORDER BY flight_id;
```

#3



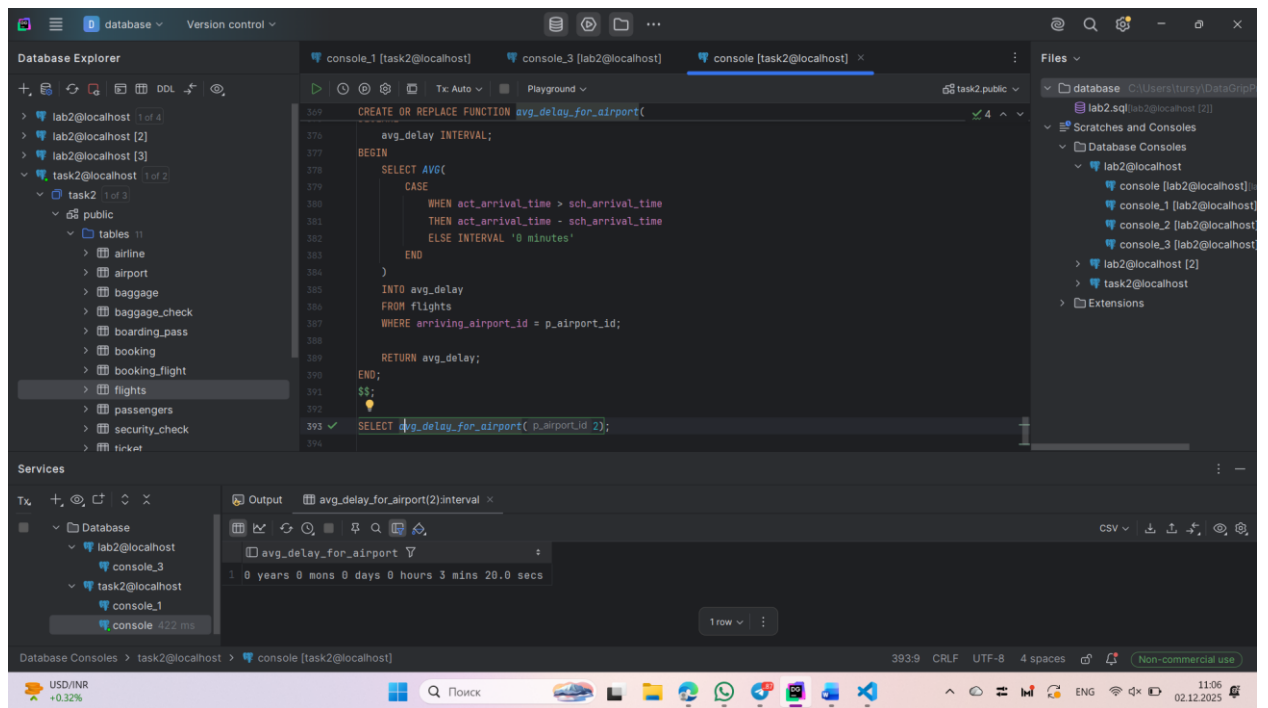
```

CREATE OR REPLACE PROCEDURE list_flights_by_airport(
    p_airport_id INT
)
LANGUAGE plpgsql
AS $$
DECLARE
    r RECORD;
BEGIN
    FOR r IN
        SELECT flight_id,
               sch_departure_time,
               sch_arrival_time,
               departing_gate,
               arriving_gate,
               airline_id,
               status
        FROM flights
        WHERE departing_airport_id = p_airport_id
        ORDER BY sch_departure_time
    LOOP
        RAISE NOTICE
        'Flight % | Departs: % | Arrives: % | Gate: % | Airline: % | Status: %',
        r.flight_id,
        r.sch_departure_time,
        r.sch_arrival_time,
        r.departing_gate,
        r.arriving_gate,
        r.airline_id,
        r.status;
    END LOOP;
END;
$$;

CALL list_flights_by_airport(1);

```

#4



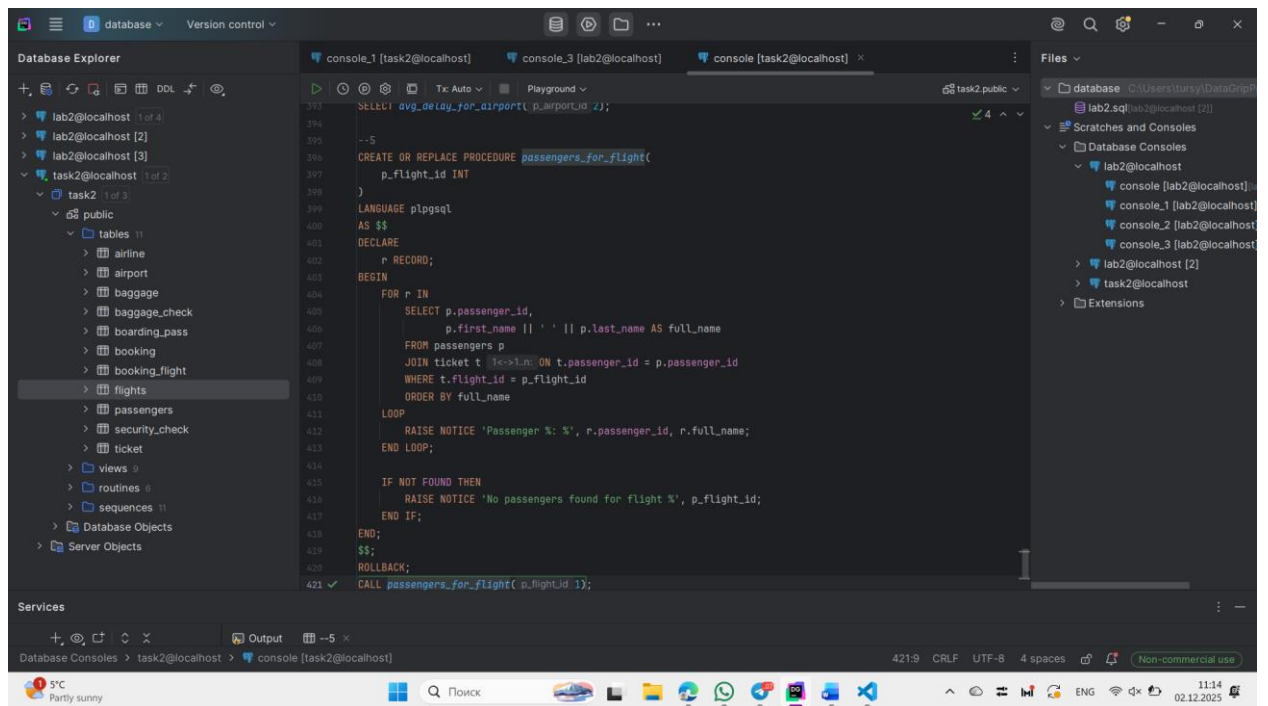
```

CREATE OR REPLACE FUNCTION avg_delay_for_airport(
    p_airport_id INT
)
RETURNS INTERVAL
LANGUAGE plpgsql
AS $$
DECLARE
    avg_delay INTERVAL;
BEGIN
    SELECT AVG(
        CASE
            WHEN act_arrival_time > sch_arrival_time
            THEN act_arrival_time - sch_arrival_time
            ELSE INTERVAL '0 minutes'
        END
    )
    INTO avg_delay
    FROM flights
    WHERE arriving_airport_id = p_airport_id;

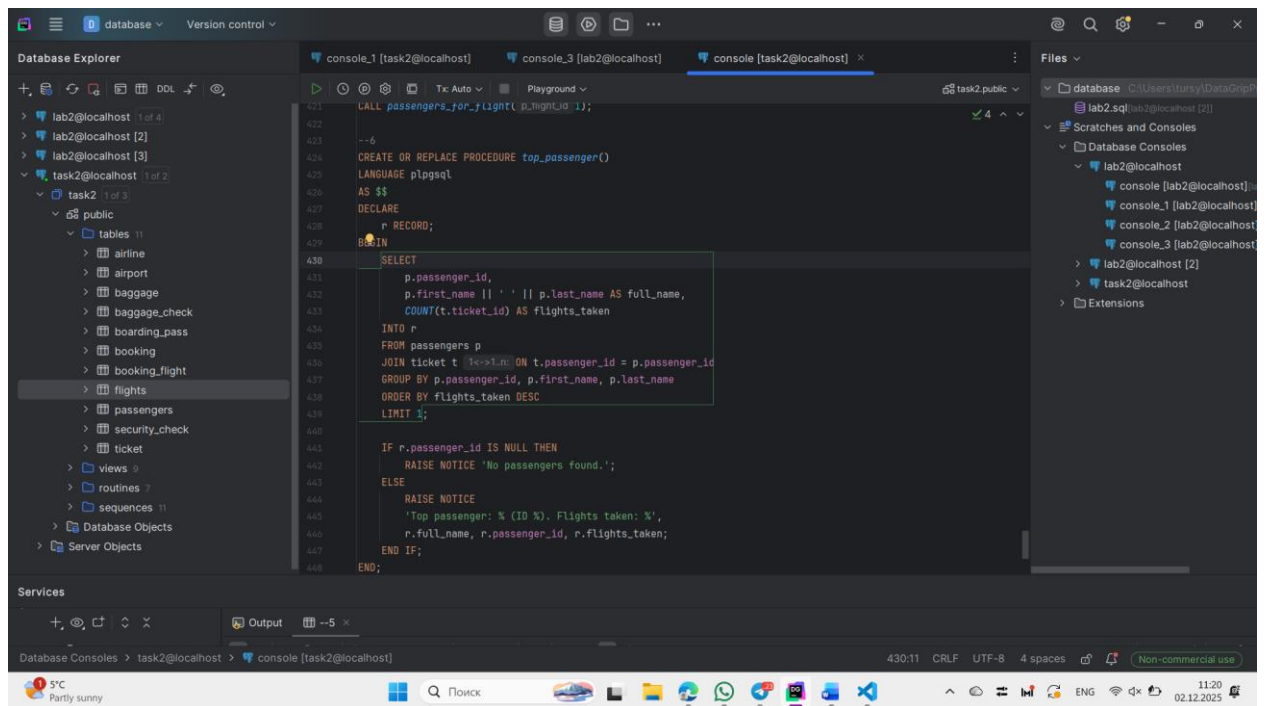
    RETURN avg_delay;
END;
$$;

SELECT avg_delay_for_airport(2);
  
```

#5



```
CREATE OR REPLACE PROCEDURE passengers_for_flight(  
    p_flight_id INT  
)  
  
LANGUAGE plpgsql  
  
AS $$  
  
DECLARE  
    r RECORD;  
  
BEGIN  
    FOR r IN  
        SELECT p.passenger_id,  
               p.first_name || ' ' || p.last_name AS full_name  
        FROM passengers p  
        JOIN tickets t ON t.passenger_id = p.passenger_id  
        WHERE t.flight_id = p_flight_id  
        ORDER BY full_name  
    LOOP  
        RAISE NOTICE 'Passenger %: %', r.passenger_id, r.full_name;  
    END LOOP;  
  
    IF NOT FOUND THEN  
        RAISE NOTICE 'No passengers found for flight %', p_flight_id;  
    END IF;  
  
END;$$  
  
#6
```



```

CREATE OR REPLACE PROCEDURE top_passenger()
LANGUAGE plpgsql
AS $$
DECLARE
    r RECORD;
BEGIN
    SELECT
        p.passenger_id,
        p.first_name || ' ' || p.last_name AS full_name,
        COUNT(t.ticket_id) AS flights_taken
    INTO r
    FROM passengers p
    JOIN ticket t ON t.passenger_id = p.passenger_id
    GROUP BY p.passenger_id, p.first_name, p.last_name
    ORDER BY flights_taken DESC
    LIMIT 1;

    IF r.passenger_id IS NULL THEN
        RAISE NOTICE 'No passengers found.';
    ELSE
        RAISE NOTICE
        'Top passenger: % (ID %). Flights taken: %',
        r.full_name, r.passenger_id, r.flights_taken;
    END IF;
END;
$$;

```

#7

Database Explorer

lab2@localhost [3]  
task2@localhost 1 of 2  
task2 1 of 3  
public  
tables 11  
airline  
airport  
baggage  
baggage\_check  
boarding\_pass  
booking  
booking\_flight  
flights  
passengers  
security\_check  
ticket  
columns 5  
keys 1  
foreign keys 2  
indexes 1  
views 9  
routines 9  
sequences 11  
Database Objects  
Server Objects

console\_1 [task2@localhost] console\_3 [lab2@localhost] console [task2@localhost]

CREATE OR REPLACE PROCEDURE flights\_delayed\_24h()  
DECLARE  
r RECORD;  
BEGIN  
FOR r IN  
SELECT  
flight\_id,  
sch\_departure\_time,  
act\_departure\_time,  
act\_departure\_time - sch\_departure\_time AS delay  
FROM flights  
WHERE act\_departure\_time - sch\_departure\_time > INTERVAL '24 hours'  
ORDER BY delay DESC  
LOOP  
RAISE NOTICE  
'Flight % | Scheduled: % | Actual: % | Delay: %',  
r.flight\_id,  
r.sch\_departure\_time,  
r.act\_departure\_time,  
r.delay;  
END LOOP;  
IF NOT FOUND THEN  
RAISE NOTICE 'No flights delayed more than 24 hours.';  
END IF;  
END;  
\$\$;  
CALL flights\_delayed\_24h();

Files

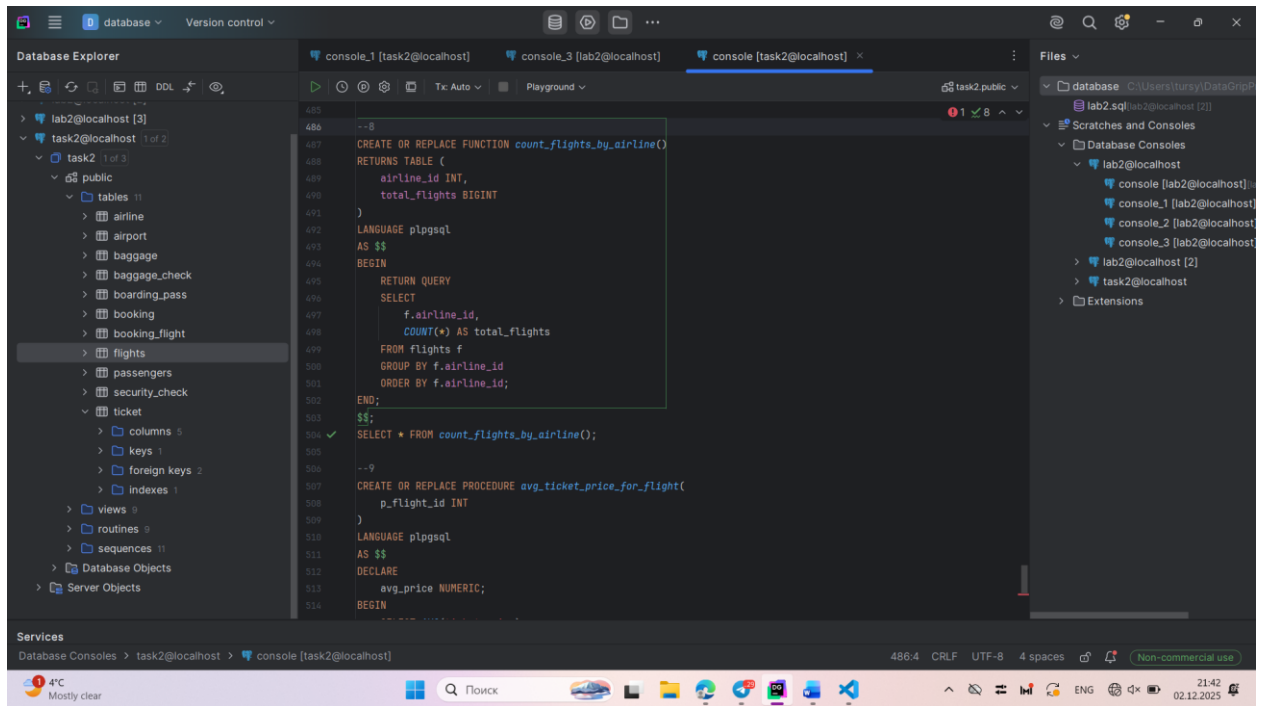
database C:\Users\turyy\DataGrid  
lab2.sql [lab2@localhost [2]]  
Scratches and Consoles  
Database Consoles  
lab2@localhost  
console [lab2@localhost]  
console\_1 [lab2@localhost]  
console\_2 [lab2@localhost]  
console\_3 [lab2@localhost]  
lab2@localhost [2]  
task2@localhost  
Extensions

Services  
Database > task2@localhost > task2 > public > tables > flights

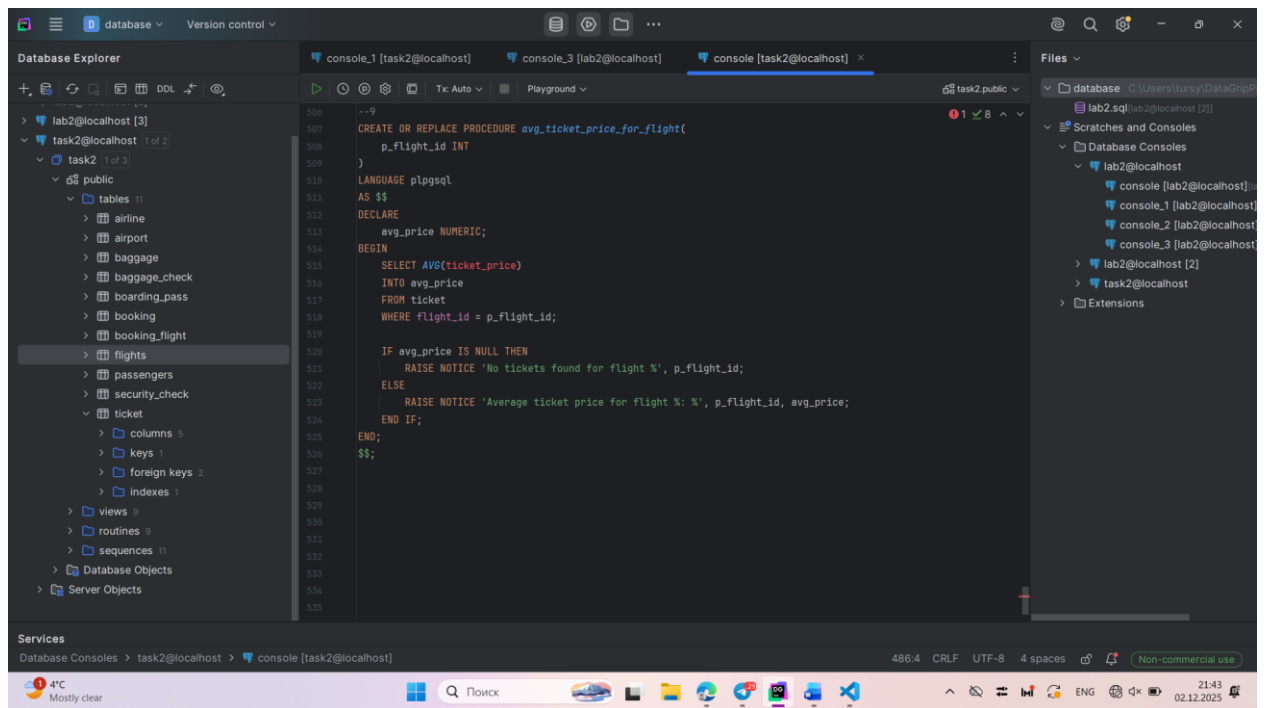
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#8



#9



#10

Database Explorer

lab2@localhost [3]  
task2@localhost 1 of 2  
task2 1 of 3  
public  
tables 11  
airline  
airport  
baggage  
baggage\_check  
boarding\_pass  
booking  
booking\_flight  
flights  
passengers  
security\_check  
ticket  
columns 5  
keys 1  
foreign keys 2  
indexes 1  
views 9  
routines 10  
sequences 11  
Database Objects  
Server Objects

console\_1 [task2@localhost] console\_3 [lab2@localhost] console [task2@localhost]

```
529 CREATE OR REPLACE PROCEDURE most_expensive_flight()
530 LANGUAGE plpgsql
531 AS $$
532 DECLARE
533     r RECORD;
534 BEGIN
535     SELECT
536         f.flight_id,
537         f.departing_airport_id,
538         f.arriving_airport_id,
539         t.price
540     INTO r
541     FROM flights f
542     JOIN ticket t ON t.flight_id = f.flight_id
543     ORDER BY t.price DESC
544     LIMIT 1;
545
546     IF r.flight_id IS NULL THEN
547         RAISE NOTICE 'No flights found.';
548     ELSE
549         RAISE NOTICE
550         'Most expensive flight: ID %, From airport %, To airport %, Price %',
551         r.flight_id,
552         r.departing_airport_id,
553         r.arriving_airport_id,
554         r.price;
555     END IF;
556 END;
```

Files

database C:\Users\turyy\DataGrid  
lab2.sql [lab2@localhost [2]]  
Scratches and Consoles  
Database Consoles  
lab2@localhost  
console [lab2@localhost]  
console\_1 [lab2@localhost]  
console\_2 [lab2@localhost]  
console\_3 [lab2@localhost]  
lab2@localhost [2]  
task2@localhost  
Extensions

Services

Database Consoles > task2@localhost > console [task2@localhost]

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